

Data Driven Analysis of Socioeconomic Determinants of Residential Solar Adoption in Pakistan

*A Comprehensive Statistical Analysis of over 200 Multi-Regional Households
An Independent Research Report*

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1. Abstract

Pakistan's residential power sector faces an unprecedented affordability crisis, driven by an aggressive 155% escalation in national grid tariffs alongside macroeconomic structural adjustments. This independent research analyzes primary empirical data from a multi-regional online survey https://docs.google.com/forms/d/1M_eKRCwqIVc_wMcWcm-PsPvC6FSjmkyp_i_bCzRF27Sw/viewanalytics of 200 households across six administrative territories to evaluate distributed solar photovoltaic (PV) adoption trends. The findings reveal that while dropping global manufacturing costs have catalyzed a rapid adoption wave among wealthy homeowners, deep systemic barriers systematically isolate lower-income tiers, apartment dwellers, and tenants. High upfront capital costs (50.0%), restricted property tenure rights (29.6%), and regulatory changes regarding net-metering policies emerge as the primary bottlenecks to widespread adoption. To prevent decentralized solar from remaining an exclusive asset that leaves vulnerable grid consumers to absorb growing utility capacity charges, Pakistan must transition from passive market observation to targeted regulatory interventions. This report outlines barriers and proposes actionable solutions for long-term regulatory policy.

2. Introduction & Background Context

The macroeconomic stability of Pakistan's power landscape has reached a financial tipping point. For decades, the power sector has been crippled by an accumulation of 'circular debt' stemming from systemic grid inefficiencies, low recovery rates, and dollar-indexed Capacity Purchase Price (CPP) contracts with Independent Power Producers (IPPs). To satisfy international fiscal consolidation targets, the state has aggressively raised utility rates, driving residential baseline tariffs up by more than 155%. Concurrently, a global supply glut in PV manufacturing has dropped component prices to historic lows. This convergence of soaring grid bills and affordable hardware has triggered a massive, market-driven transition toward decentralized consumer generation. As of 31 March 2025, Pakistan has reached over ~2,813 MW of net-metered rooftop solar installed [4] of distributed solar integrated into the national grid via net-metering registries.

However, this unguided transition exposes deep structural inequities. Under the current distributed generation paradigm, solar adoption is highly regressive, creating severe financial risks for the non-solar consumer base. As high-income households invest in solar arrays to eliminate their billing brackets, wealthier consumers effectively withdraw their financial support from the public utility framework. Because fixed capacity charges are redistributed across fewer remaining consumed grid units, power distribution companies (DISCOs) raise base tariffs further. This shifts the financial burden of grid maintenance entirely onto lower-income families who lack the capital to defect. Furthermore, tenant populations and apartment dwellers are physically and

contractually excluded due to a misalignment of incentives with landlords and fractured rooftop property rights.

3. Research Methodology & Demographics

This study utilizes primary empirical data compiled from an independent multi-regional survey of over 200 households across Pakistan. The questionnaire was deployed digitally, capturing data on geographical location, residential tenure, household size, monthly income distribution, adoption status, structural obstacles, and primary operational drivers. Respondents were permitted to select multiple variables for causation and barrier questions to accurately capture household decision-making dynamics.

3.1 Complete Sample Demographics

The table below details the socioeconomic and geographical distribution of the surveyed households.

Demographic Variable	Sample Breakdown (Category)	Sample Share (%) / Count (n)
Geographical Region	Punjab	39.9% (n=81)
	Sindh	26.1% (n=53)
	Islamabad Capital Territory (ICT)	19.2% (n=39)
	Khyber Pakhtunkhwa (KPK)	7.4% (n=15)
	Azad Jammu & Kashmir (AJK)	3.0% (n=6)
	Gilgit-Baltistan (GB)	2.5% (n=5)
	Balochistan	2.0% (n=4)
Monthly Household Income	Below 50,000 PKR	1.5% (n=3)
	50,000 - 100,000 PKR	13.5% (n=27)
	100,000 - 200,000 PKR	20.9% (n=42)
	200,000 - 300,000 PKR	23.4% (n=47)
	300,000 - 400,000 PKR	17.9% (n=36)
	Above 500,000 PKR	22.9% (n=46)
Property Tenure Type	Owned House	79.3% (n=159)
	Rented House	15.8% (n=31)
	Apartment	3.9% (n=8)
	Other Property Types	1.0% (n=2)

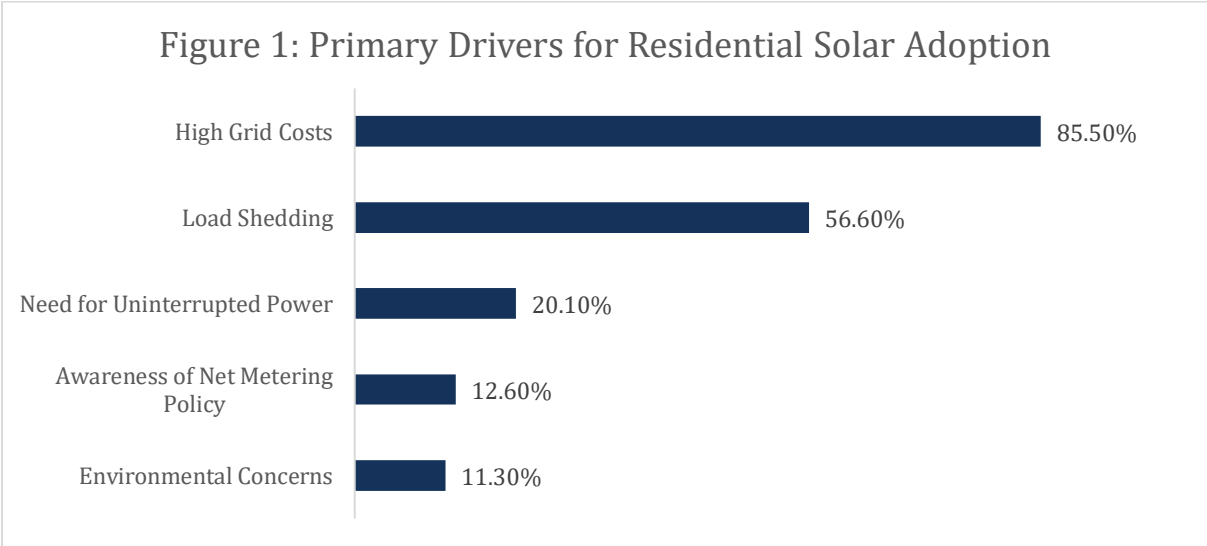
Table 1. Response of online survey

(<https://docs.google.com/spreadsheets/d/1JSWD8c9IISTCw0KNdrJVJvGPRoGLICGTuGklptHI0/edit?usp=sharing>)

4. Data Analysis & Empirical Findings

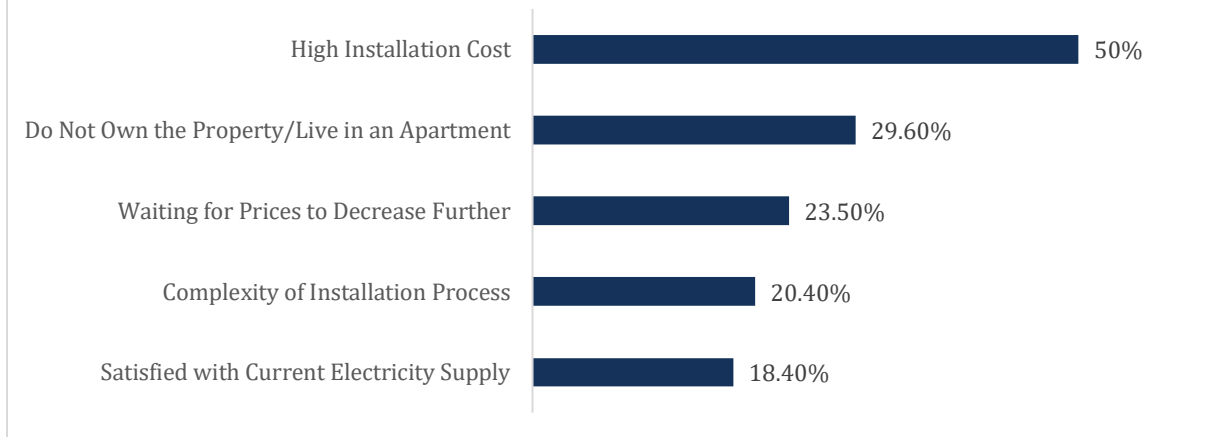
The cross-tabulated dataset reveals distinct socio-demographic fractures in Pakistan's distributed renewable energy market, separating consumer intent from actual economic and infrastructural.

Tariff-Driven Grid Defection: Economic preservation is the overwhelming catalyst for solar adoption. An immense 85.5% of adopters cited high grid electricity costs as their primary driver for installing solar, vastly outperforming grid unreliability/load-shedding (56.6%), the need for uninterrupted power (20.1%), awareness of net metering policy (12.6%) and environmental concerns (11.3%). This empirical reality indicates that aggressive public utility tariff hikes act as a regressive economic penalty, forcing financially viable consumers off the national grid via Utility Financial Defection.



Liquidity Constraints and Credit Deficiencies: Upfront capital expenditure remains the absolute bottleneck for non-adopters, even amid falling global panel prices. Exactly 50.0% of non-adopters identified high installation costs as their primary barrier. This delay is worsened by market speculation, with 23.5% deferring adoption to wait for further price drops. Institutional financing has failed to bridge this gap: only 4.4% of solar adopters utilized formal bank loans, proving that distributed solar adoption in Pakistan is strictly limited to cash-liquid households.

Figure 2: Socio-Economic Barriers to Residential Solar Adoption



Tenure and Property Constraints: Current net-metering regulations completely exclude non-homeowners. While homeownership dominates the total sample, 29.6% of non-adopters stated they cannot install solar because they live in rented properties or apartments, while 19.4% lack sufficient rooftop space. Because tenants (15.8% of sample) and apartment residents (3.9%) have no legal framework to divide solar generation costs or credits with landlords, they are structurally trapped under inflationary grid tariffs.

5. Discussion & Policy Recommendations

Based strictly on the empirical bottlenecks identified within the household survey, Pakistan's energy regulators and financial institutions must transition from passive market observation to targeted structural interventions:

1. Launch On-Bill 'Pay-As-You-Save' (PAYS) Microfinancing: To address the 50.0% upfront capital barrier impacting middle-income tiers (100k–400k PKR/month), the State Bank of Pakistan (SBP) must restructure its green banking guidelines. Because only 4.4% of consumers currently access formal solar loans, commercial banks should partner with power distribution companies (DISCOs) to deploy an on-bill financing framework. Under this model, tier-1 solar hardware is installed with zero upfront capital from the consumer. The installation cost is amortized directly onto their monthly utility bill and fully offset by the immediate generation savings, removing the liquidity constraint for middle-class households.

2. Establish a Legal Virtual Net-Metering (VNM) Framework: NEPRA must urgently amend distributed generation regulations to permit Virtual Net-Metering to address the 29.6% tenant and apartment barrier. This regulatory update will allow multi-family residential buildings or landlords of rented properties to install a single, shared solar

array on available rooftop space. The net-metering credits generated can then be digitally distributed and credited across separate utility accounts. This legal framework resolves the landlord-tenant incentive misalignment and extends solar access to the 19.4% of urban consumers facing rooftop space constraints.

3. Implement Net-Metering Policy Safeguards to Stabilize Demand: To counter the 23.5% of consumers delaying adoption due to price speculation and the 17.3% paralyzed by regulatory uncertainty, the Ministry of Energy must establish policy consistency. The government should legally guarantee that any residential net-metering contract signed under current buy-back rates is grandfathered in for a fixed, minimum 10-year term. Providing explicit regulatory stability will protect consumer investments, prevent artificial market stalls from deflationary waiting, and create predictable long-term adoption rates.

6. References

[1] National Electric Power Regulatory Authority (NEPRA). State of Industry Report 2024: Distributed Generation and Residential Net-Metering Integration Metrics. Islamabad, Pakistan.

[2] State Bank of Pakistan (SBP). Revised Financing Scheme for Renewable Energy (IH&SME Finance Circulars). Karachi, Pakistan.

[3] Pakistan Institute of Development Economics (PIDE). From Net Metering to Net Billing: Tariff Design, Distribution Sector Stress, and Regressive Cost-Shifting in Pakistan. Research Series, 2026.

[4] [Solar power in Pakistan - Wikipedia](#)